Solution

(a) If $\hat{A}$ and $\hat{B}$ are Hermitian, we can write

$$[\hat{A},\ \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A},\ \hat{B}]; \tag{2.93}$$

that is, the commutator of $\hat{A}$ and $\hat{B}$ is anti-Hermitian: $[\hat{A},\ \hat{B}]^\dagger = -[\hat{A},\ \hat{B}]$.

(b) Using the distributivity relation (2.87), we have

$$\begin{aligned}[\hat{A},\ [\hat{B},\ \hat{C}]\hat{D}] &= [\hat{B},\ \hat{C}][\hat{A},\ \hat{D}] + [\hat{A},\ [\hat{B},\ \hat{C}]]\hat{D} \\ &= (\hat{B}\hat{C} - \hat{C}\hat{B})(\hat{A}\hat{D} - \hat{D}\hat{A}) + \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B})\hat{D} - (\hat{B}\hat{C} - \hat{C}\hat{B})\hat{A}\hat{D} \\ &= \hat{C}\hat{B}\hat{D}\hat{A} - \hat{B}\hat{C}\hat{D}\hat{A} + \hat{A}\hat{B}\hat{C}\hat{D} - \hat{A}\hat{C}\hat{B}\hat{D}. \end{aligned} \tag{2.94}$$

2.4.5 Uncertainty Relation between Two Operators

An interesting application of the commutator algebra is to derive a general relation giving the uncertainties product of two operators, $\hat{A}$ and $\hat{B}$. In particular, we want to give a formal derivation of Heisenberg's uncertainty relations.

Let $\langle\hat{A}\rangle$ and $\langle\hat{B}\rangle$ denote the expectation values of two Hermitian operators $\hat{A}$ and $\hat{B}$ with respect to a normalized state vector $|\ \psi\rangle$: $\langle\hat{A}\rangle = \langle\psi\ |\ \hat{A}\ |\ \psi\rangle$ and $\langle\hat{B}\rangle = \langle\psi\ |\ \hat{B}\ |\ \psi\rangle$. Introducing the operators $\Delta\hat{A}$ and $\Delta\hat{B}$,

$$\Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle, \qquad \Delta\hat{B} = \hat{B} - \langle\hat{B}\rangle, \tag{2.95}$$

we have $(\Delta\hat{A})^2 = \hat{A}^2 - 2\hat{A}\langle\hat{A}\rangle + \langle\hat{A}\rangle^2$ and $(\Delta\hat{B})^2 = \hat{B}^2 - 2\hat{B}\langle\hat{B}\rangle + \langle\hat{B}\rangle^2$, and hence

$$\langle\psi\ |\ (\Delta\hat{A})^2\ |\ \psi\rangle = \langle(\Delta\hat{A})^2\rangle = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2, \qquad \langle(\Delta\hat{B})^2\rangle = \langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2, \tag{2.96}$$

where $\langle\hat{A}^2\rangle = \langle\psi\ |\ \hat{A}^2\ |\ \psi\rangle$ and $\langle\hat{B}^2\rangle = \langle\psi\ |\ \hat{B}^2\ |\ \psi\rangle$. The *uncertainties* ΔA and ΔB are defined by

$$\boxed{\Delta A = \sqrt{\langle(\Delta\hat{A})^2\rangle} = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2}, \qquad \Delta B = \sqrt{\langle(\Delta\hat{B})^2\rangle} = \sqrt{\langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2}.} \tag{2.97}$$

Let us write the action of the operators (2.95) on any state $|\ \psi\rangle$ as follows:

$$|\ \chi\rangle = \Delta\hat{A}\ |\ \psi\rangle = \left(\hat{A} - \langle\hat{A}\rangle\right)|\ \psi\rangle, \qquad |\ \phi\rangle = \Delta\hat{B}\ |\ \psi\rangle = \left(\hat{B} - \langle\hat{B}\rangle\right)|\ \psi\rangle. \tag{2.98}$$

The Schwarz inequality for the states $|\ \chi\rangle$ and $|\ \phi\rangle$ is given by

$$\langle\chi\ |\ \chi\rangle\langle\phi\ |\ \phi\rangle \geq |\langle\chi\ |\ \phi\rangle|^2. \tag{2.99}$$

Since $\hat{A}$ and $\hat{B}$ are Hermitian, $\Delta\hat{A}$ and $\Delta\hat{B}$ must also be Hermitian: $\Delta\hat{A}^\dagger = \hat{A}^\dagger - \langle\hat{A}\rangle = \hat{A} - \langle\hat{A}\rangle = \Delta\hat{A}$ and $\Delta\hat{B}^\dagger = \hat{B} - \langle\hat{B}\rangle = \Delta\hat{B}$. Thus, we can show the following three relations:

$$\langle\chi\ |\ \chi\rangle = \langle\psi\ |\ (\Delta\hat{A})^2\ |\ \psi\rangle, \quad \langle\phi\ |\ \phi\rangle = \langle\psi\ |\ (\Delta\hat{B})^2\ |\ \psi\rangle, \quad \langle\chi\ |\ \phi\rangle = \langle\psi\ |\ \Delta\hat{A}\Delta\hat{B}\ |\ \psi\rangle. \tag{2.100}$$

For instance, since $\Delta\hat{A}^\dagger = \Delta\hat{A}$ we have $\langle\chi \mid \chi\rangle = \langle\psi \mid \Delta\hat{A}^\dagger\Delta\hat{A} \mid \psi\rangle = \langle\psi \mid (\Delta\hat{A})^2 \mid \psi\rangle = \langle(\Delta\hat{A})^2\rangle$. Hence, the Schwarz inequality (2.99) becomes

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \;\geq\; \left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2. \tag{2.101}$$

Notice that the last term $\Delta\hat{A}\Delta\hat{B}$ of this equation can be written as

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}[\Delta\hat{A},\ \Delta\hat{B}] + \frac{1}{2}\{\Delta\hat{A},\ \Delta\hat{B}\} = \frac{1}{2}[\hat{A},\ \hat{B}] + \frac{1}{2}\{\Delta\hat{A},\ \Delta\hat{B}\}, \tag{2.102}$$

where we have used the fact that $[\Delta\hat{A},\ \Delta\hat{B}] = [\hat{A},\ \hat{B}]$. Since $[\hat{A},\ \hat{B}]$ is anti-Hermitian and $\{\Delta\hat{A},\ \Delta\hat{B}\}$ is Hermitian and since the expectation value of a Hermitian operator is real and that the expectation value of an anti-Hermitian operator is imaginary (see Example 2.6), the expectation value $\langle\Delta\hat{A}\Delta\hat{B}\rangle$ of (2.102) becomes equal to the sum of a real part $\langle\{\Delta\hat{A},\ \Delta\hat{B}\}\rangle/2$ and an imaginary part $\langle[\hat{A},\ \hat{B}]\rangle/2$; hence

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 = \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2 + \frac{1}{4}\left|\langle\{\Delta\hat{A},\ \Delta\hat{B}\}\rangle\right|^2. \tag{2.103}$$

Since the last term is a positive real number, we can infer the following relation:

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \;\geq\; \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2. \tag{2.104}$$

Comparing equations (2.101) and (2.104), we conclude that

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \;\geq\; \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2, \tag{2.105}$$

which (by taking its square root) can be reduced to

$$\boxed{\Delta A\Delta B \geq \frac{1}{2}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|.} \tag{2.106}$$

This uncertainty relation plays an important role in the formalism of quantum mechanics. Its application to position and momentum operators leads to the Heisenberg uncertainty relations, which represent one of the cornerstones of quantum mechanics; see the next example.

Example 2.9 (Heisenberg uncertainty relations)
Find the uncertainty relations between the components of the position and the momentum operators.

Solution
By applying (2.106) to the x-components of the position operator $\hat{X}$, and the momentum operator $\hat{P}_x$, we obtain $\Delta x\,\Delta p_x \;\geq\; \frac{1}{2} \mid \langle[\hat{X},\ \hat{P}_x]\rangle \mid$. But since $[\hat{X},\ \hat{P}_x] = i\hbar\hat{I}$, we have $\Delta x\,\Delta p_x \geq \hbar/2$; the uncertainty relations for the $y-$ and $z-$ components follow immediately:

$$\boxed{\Delta x\,\Delta p_x \geq \frac{\hbar}{2}, \qquad \Delta y\,\Delta p_y \geq \frac{\hbar}{2}, \qquad \Delta z\,\Delta p_z \geq \frac{\hbar}{2}.} \tag{2.107}$$

These are the Heisenberg uncertainty relations.
